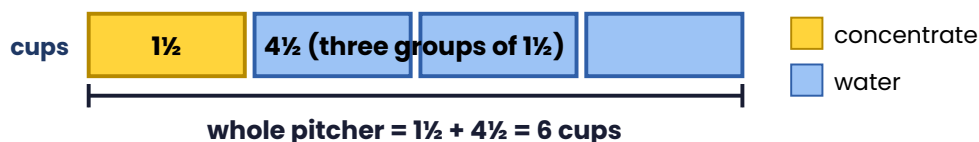


PART-TO-PART & PART-TO-WHOLE

*Add the parts.
Scale them all.*

A **ratio** compares two quantities. In a mixture, you can compare one **part** to another part, or one part to the **whole**. Real life asks you to switch between them — and to scale by fractions and mixed numbers.

EXAMPLE · LEMONADE



PART : PART

$$1\frac{1}{2} : 4\frac{1}{2} = 1 : 3$$

concentrate : water
Compares one part to the **other part**.

PART : WHOLE

$$1\frac{1}{2} : 6 = 1 : 4$$

concentrate : whole pitcher
Compares one part to the **total**.

CROSS BETWEEN THEM

part : part → **part : whole** $a : b \rightarrow a : (a + b)$ *add the parts*

part : whole → **part : part** $a : w \rightarrow a : (w - a)$ *subtract to find the other part*

Same ratio? $a : b$ and $c : d$ are equivalent when $a \times d = b \times c$ (cross products).

SCALE BY A MIXED NUMBER

Fill a **10-cup** pitcher: scale factor = $10 \div 6 = 1\frac{2}{3}$

concentrate = $1\frac{1}{2} \times 1\frac{2}{3} = 2\frac{1}{2}$ cups · water = $4\frac{1}{2} \times 1\frac{2}{3} = 7\frac{1}{2}$ cups · check: $2\frac{1}{2} + 7\frac{1}{2} = 10$ ✓

Multiply every part of the mixed number: $4 \times 2\frac{1}{2} = (4 \times 2) + (4 \times \frac{1}{2}) = 10$, **not** $8\frac{1}{2}$.

REAL WORLD CONNECTIONS

- Recipes & drinks (ingredient : ingredient vs ingredient : batch)
- Concrete & mortar (1 cement : $2\frac{1}{2}$ sand; cement : wheelbarrow)
- Paint colors ($2\frac{1}{4}$ qt blue : $\frac{3}{4}$ qt yellow → 5 qt green)
- Budgets (save : spend vs save : paycheck)
- Sports stats (made : missed vs made : attempts)
- Class data (girls : boys vs girls : whole class)



PARTS MAKE THE WHOLE.
SCALE THEM ALL!

**MATH
OPENS
DOORS!**